We define the necessary variables in this next code snippet and create our table. Notice that variables passed from the setup file are in Python and SQL, meaning they are unavailable in Scala. Although all three languages can be used in a notebook, they do not share constants or variable values between them. As a result, we define the variables we need in Scala for access in the Scala notebook. We set the volume location for files, output paths, and the `inputTable` name:

```
// variables passed from the setup file are in python
val table_name = "transaction_count_ft"
val history_table_name = "transaction_count_history"
val volumePath = "/Volumes/ml_in_action/synthetic_transactions/files/"
val outputPath = f"$volumePath/$table_name/streaming_outputs/"
val outputPath2 = f"$volumePath/$history_table_name/streaming_outputs/"
val inputTable = "ml_in_action.synthetic_transactions.raw_transactions"
```

Figure 5.12 – Setting variables, constants, and paths in Scala

Notice that we enable the CDF on our streaming feature table, `transaction_count_ft`, in *Figure 5.13*. We could publish this table to an online store if desired. Additionally, we set the table name we want to write all transactions, `transaction_count_history`:

```
sql(f"""CREATE OR REPLACE TABLE $table_name (CustomerID Int, transactionCount Int,
eventTimestamp Timestamp, isTimeout Boolean)""")
sql(f"""ALTER TABLE $table_name ALTER COLUMN CustomerID SET NOT NULL""")
sql(f"""ALTER TABLE $table_name ADD PRIMARY KEY(CustomerID)""")
sql(f"ALTER TABLE $table_name SET TBLPROPERTIES (delta.enableChangeDataFeed=true)")
```

Figure 5.13 – Creating a table and enabling CDF

Next, the `windowMinutes` constant is the number of minutes we want to aggregate transactions for each customer, whereas `maxWaitMinutes` is exactly what it sounds like. It is the minutes the stream waits for a transaction before writing out the state without new transactions. The value of `maxWaitMinutes` should always be less than the value of `windowMinutes`:

```
// aggregate transactions for windowMinutes
val windowMinutes = 2
// wait at most max_wait_minutes before writing a record
val maxWaitMinutes = 1
```

Figure 5.14 – Setting windowMinutes and maxWaitMinutes for our stream

We will initiate `FeatureEngineeringClient` and set feature table tags so we can easily see which project these tables are associated with: